

INTRINSIC SEMICONDUCTOR

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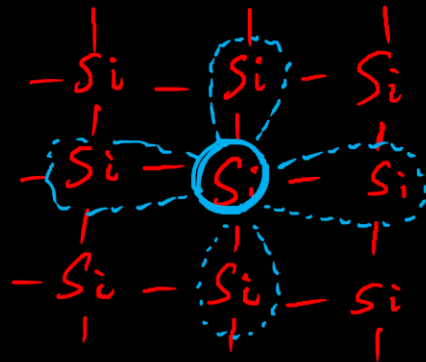
SEMI-CONDUCTORS

→ Si, Ge, GaAs, InP, N-TYPE and P-TYPE.

→ Group 14 Element-

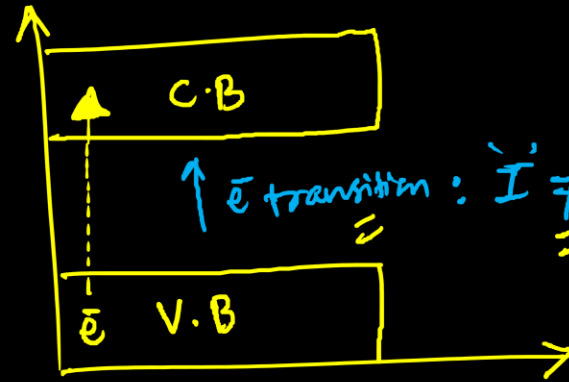
→ Valency: 4 (Outermost shell 4 e⁻s).

→ Tetravalent Element-

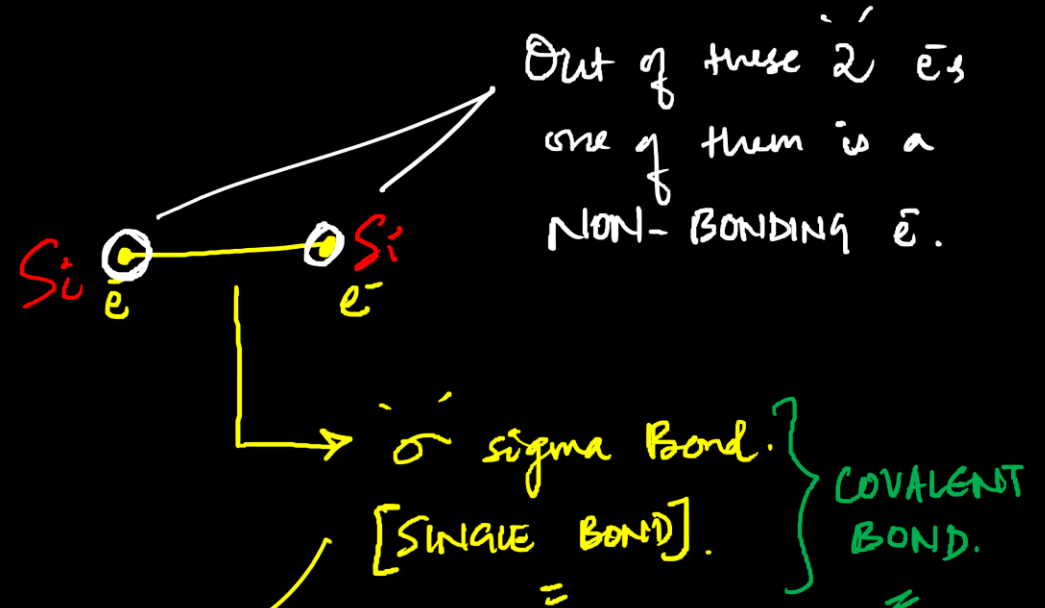


→ TETRAVALENT
ATOM

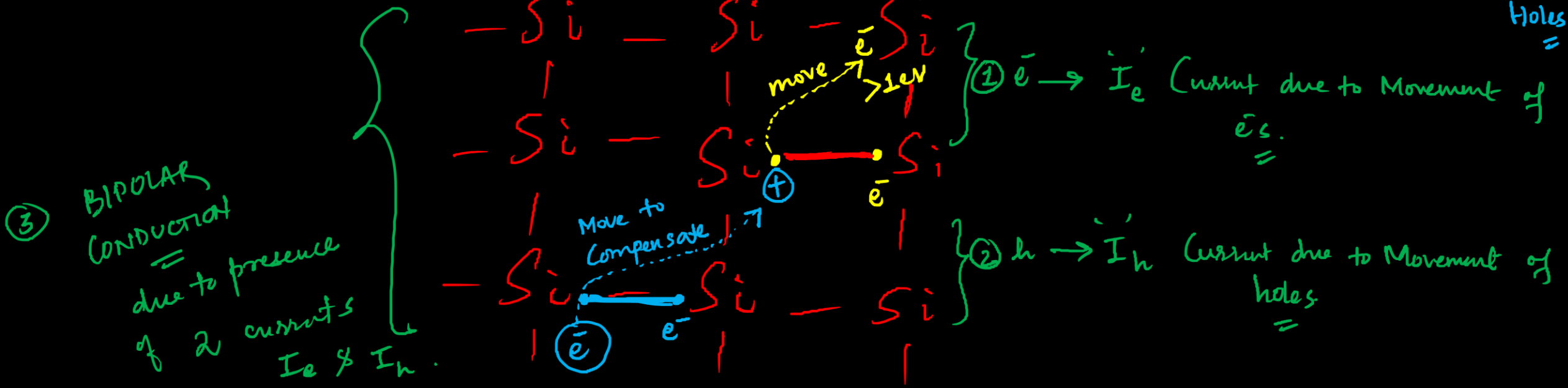
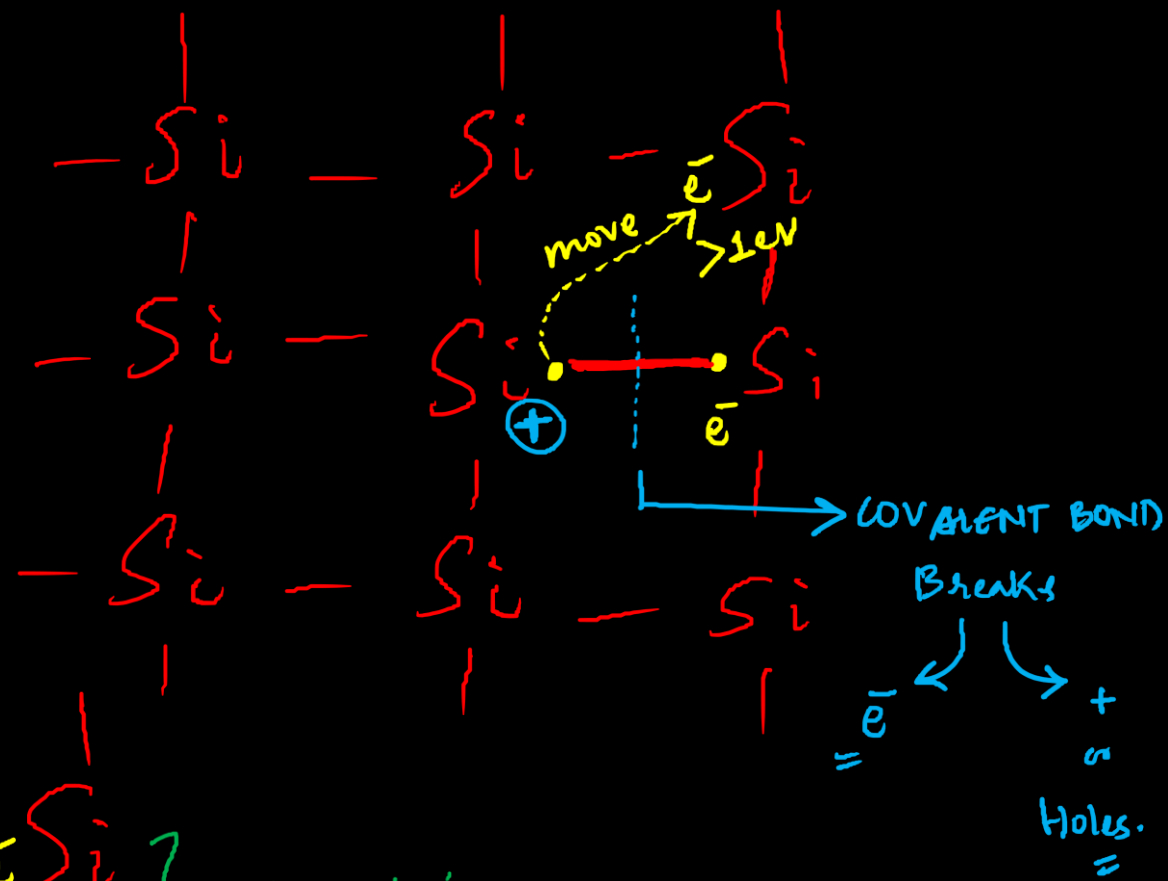
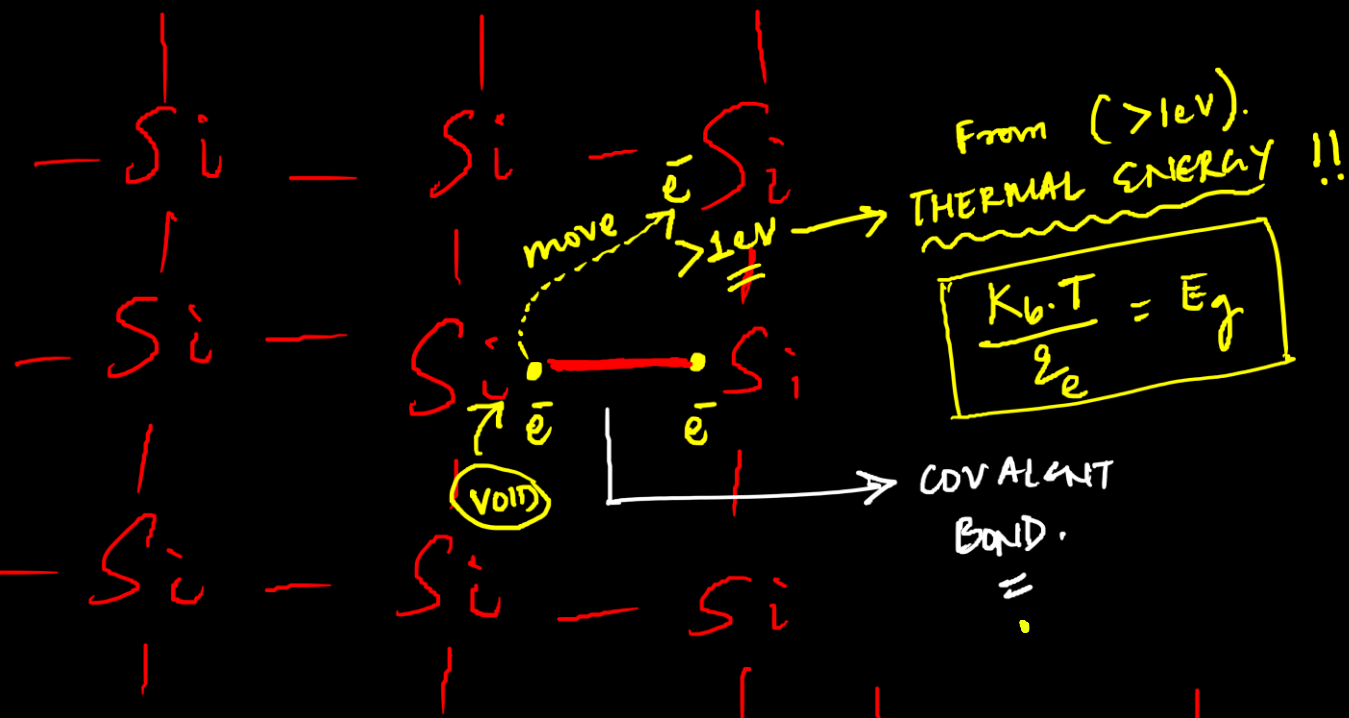
$$E_g = 1.12 \text{ eV} > 1 \text{ eV}$$



↑ e⁻ transition: $I \neq 0 \checkmark \rightarrow E_g \approx 25.87 \text{ mV} \approx \boxed{26 \text{ mV.}}$

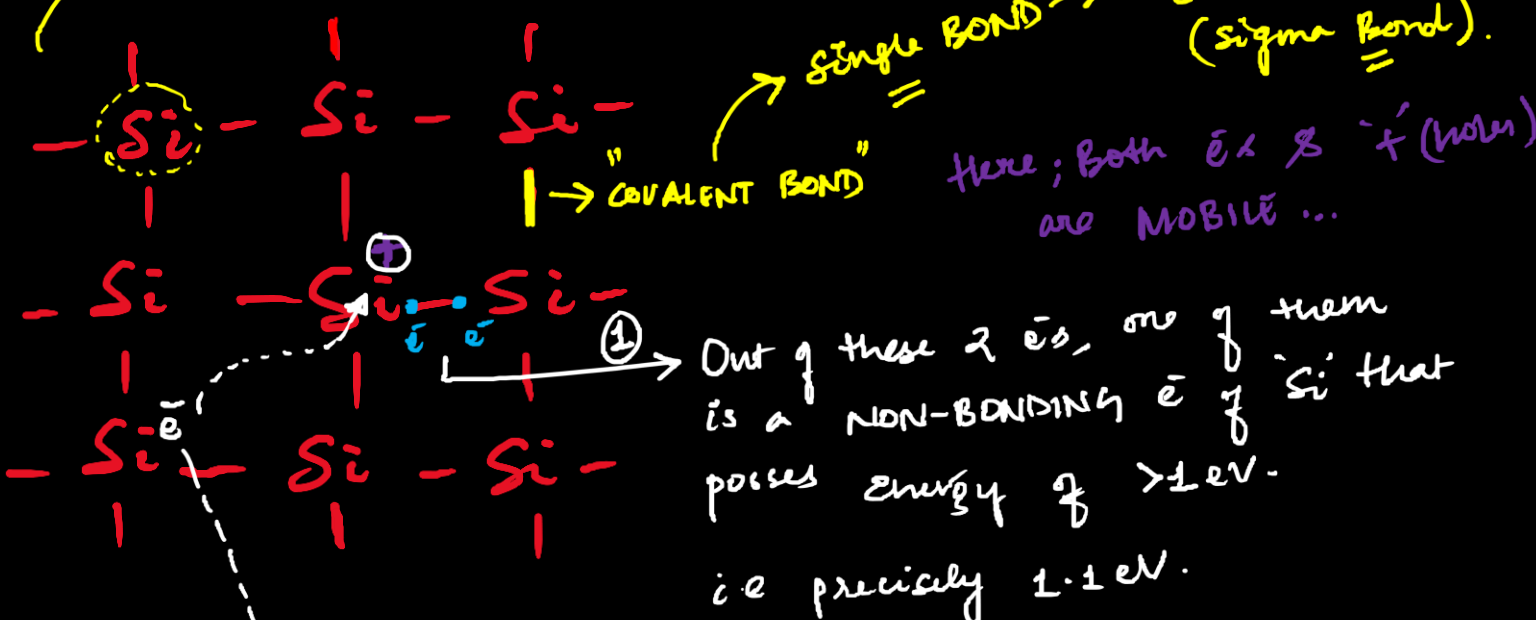


• Mechanically Strong
• Electrically Weak



③ BIPOLAR CONDUCTION = due to presence of 2 currents I_e & I_h .

Group 14 Element: 4 Valence/4



NOTE # $\rightarrow$ The Movement of e^- & h^+ charge direction is opposite to Each other =

Hence; it's [BIPOLAR CONDUCTION].

* In CONDUCTORS; there is no h^+ @ holes, etc.

$\checkmark$ ONLY e^- ... hence [UNIPOLAR CONDUCTION]

Hence; It Moves from the lattice structure of Si $\rightarrow$ CB.

④ Now; $e^- \rightarrow$ will move to fill that VACANCY ...

② Now; due to this; e^- DEFICIENT; Silicon Attains a $+ve$ charge on its head.

③ Now; to compensate this $[+ve \text{ charge}]$, Adjacent @ Neighbouring e^- come Nearby.

• Energy is nothing But Thermal Energy OR Vibrational Energy.

and; then this process will continue ...

[Breakage of COVALENT BONDS]

• 1 CB $\approx$ $1e^-$ & $1h^+$ pair.

So;

MOVEMENT $\leftarrow e^- \xrightarrow{I_{n_e}}$

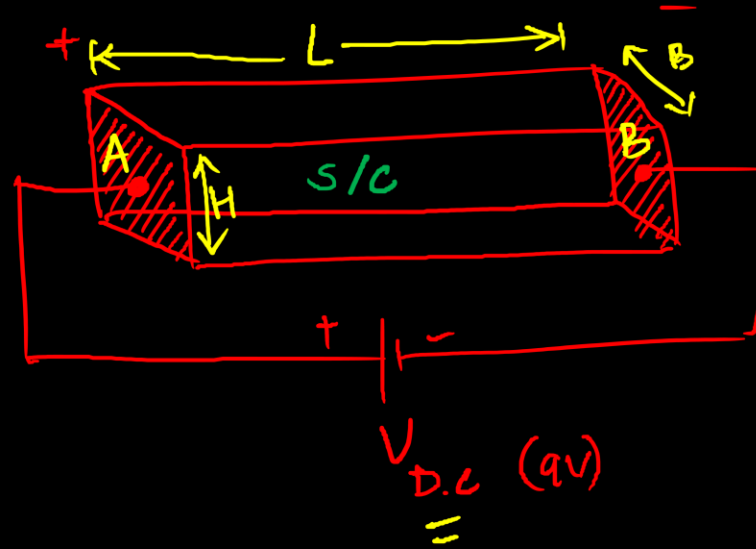
MOVEMENT $\leftarrow h^+ \xrightarrow{I_{h_p}}$

Net Current in Sustaining s/c. = $I = I_{n_e} + I_{h_p}$ ✓

Thermally Generated Current =

$$\sigma = n \cdot e \cdot \mu_e; \text{Conductors}$$

$$\sigma = n e \mu_h + n e \mu_e; \text{S/C}$$



$$R = \rho \cdot \frac{L}{A} = \frac{1}{\sigma} \cdot \frac{L}{A}$$

Conductors

$$R = \frac{1}{n e \mu_e} \cdot \frac{L}{A}$$

cond
e

S/C

$$R = \frac{1}{n e \mu_h + n e \mu_e} \cdot \frac{L}{A}$$

① μ = Mobility of $\bar{e}$ s or holes

$$= \frac{\vec{V}_d}{\vec{E}} = \frac{\text{drift velocity}}{\text{Electric field}}$$

$$\rightarrow I_{\text{total}} = I_e + I_h \quad **$$

So; $[CB \text{ breaking}] \rightleftharpoons e^- + e^- \text{ deficient site (HOLE)}$

Simultaneous process.

$e^- + h^+ \rightarrow \text{recombine} \rightarrow \text{BOND}$

This phenomenon is REVERSIBLE ...

when $T \uparrow$; Carrier conc. $\uparrow$

$[n = n_i = n_p]$

A material-dependent constant that incorporates factors like effective masses of electrons and holes, and the density of states in the conduction and valence bands. The value of A varies for different semiconductor materials.

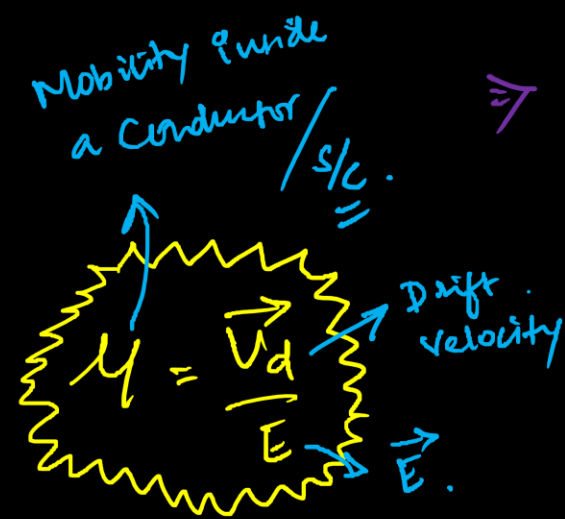
Intrinsic Carrier conc. $n_i = A \cdot T^{3/2} \cdot e^{-\frac{E_g}{KT}}$

$\rightarrow$ Forbidden Band gap energy

$\rightarrow$ Absolute Temp

$[In \text{ KELVIN Scale}]$

$1.38 \times 10^{-23} \frac{J}{K}$



$n_i^2 = A^2 \cdot T^3$

$n_i^2 \propto T^3$

At a Given const. Temperature ...

$n_p = n_i^2 = \text{const} \dots [Law \text{ of } \underline{MASS \text{ ACTION}}]$

$1.45 \times 10^{10} \text{ cm}^{-3}$

(2) For Semi-Conductors

$\sigma = n \cdot e \cdot \mu_e$

For Metals — (1)

Unipolar conduction.

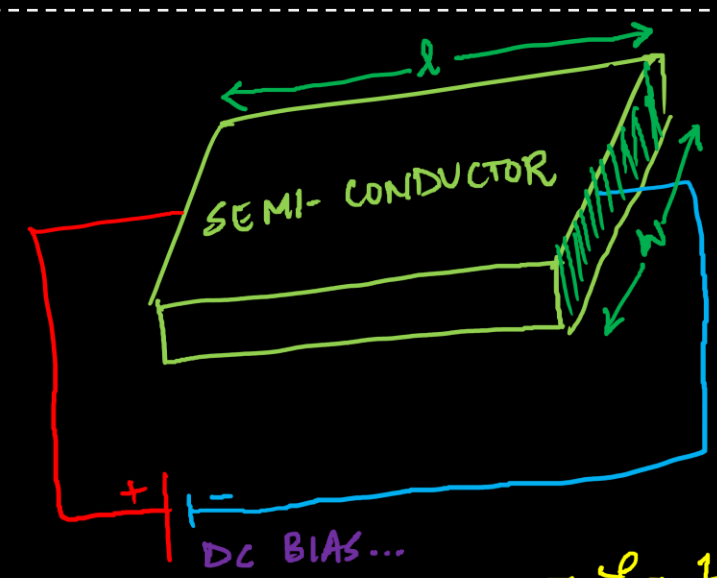
Conductivity.

No. of e^- per unit volume.

Mobility of e^- .

$\sigma = n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h$

Bipolar conduction



Resistance: $R = \rho \cdot \frac{l}{A} = \frac{1}{\sigma} \cdot \frac{l}{A}$

$\rho = \frac{1}{\sigma}$ = Conductivity.

$R = \frac{1}{n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h} \cdot \frac{l}{A}$